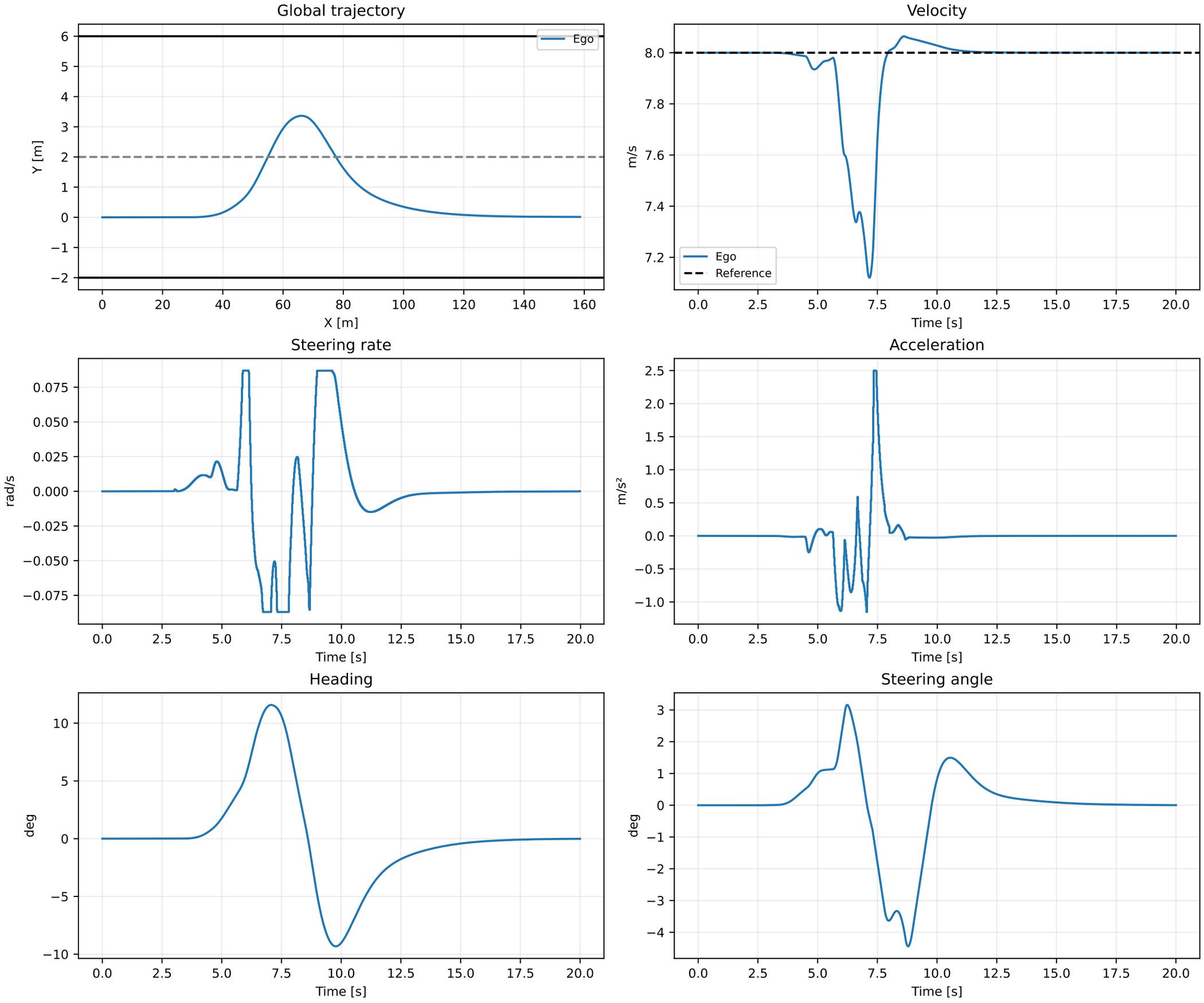
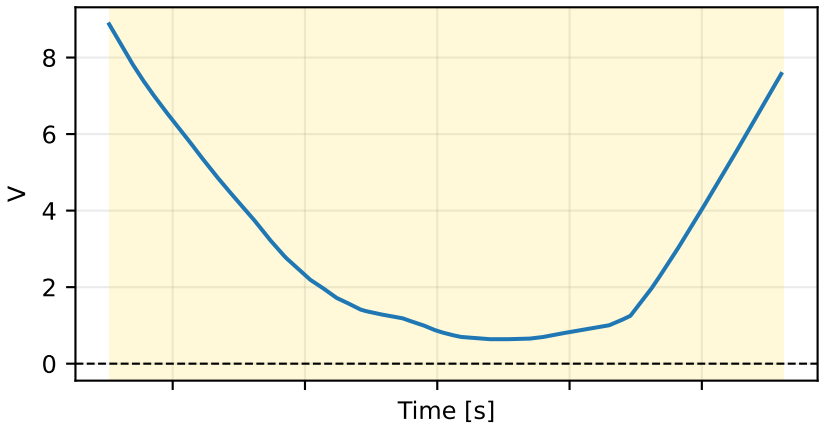
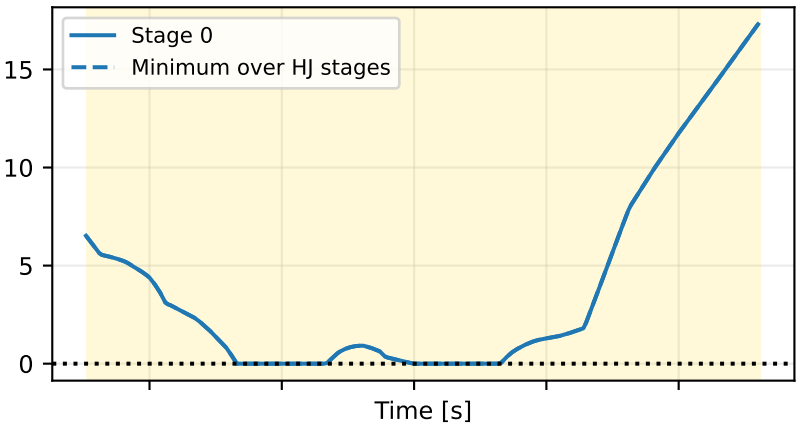

Global trajectory
Ego
Y [m]
X [m]
Velocity
m/s
Ego
Reference
Time [s]
Steering rate
rad/s
Time [s]
Acceleration
m/s²
Time [s]
Heading
deg
Time [s]
Steering angle
deg
Time [s]

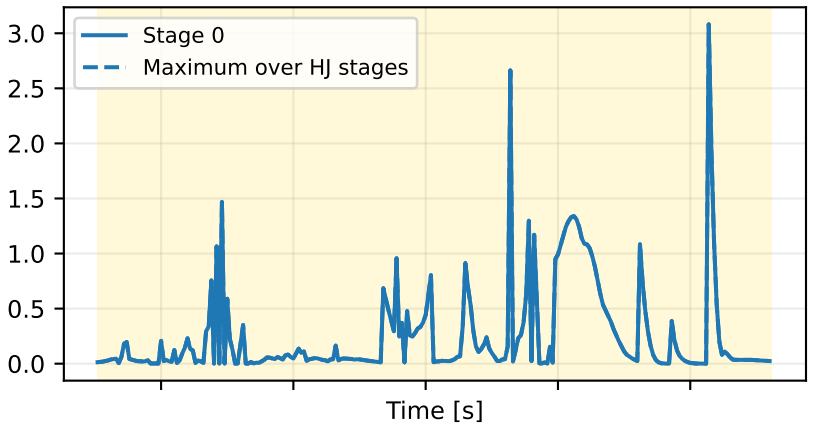
euclidean: BRT



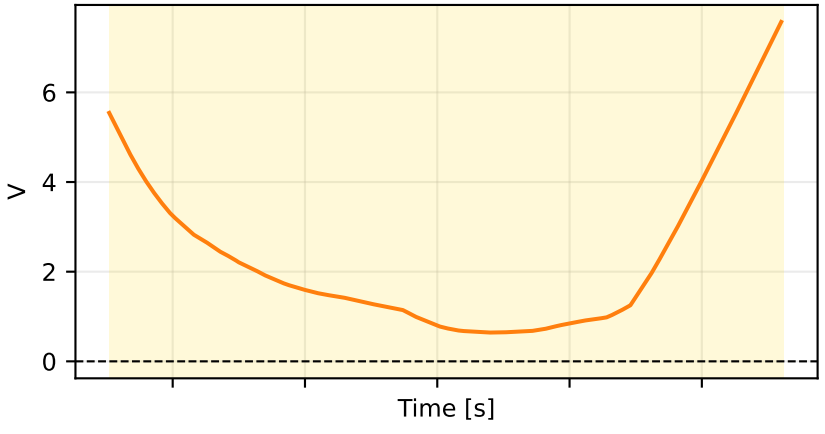
HJ residual



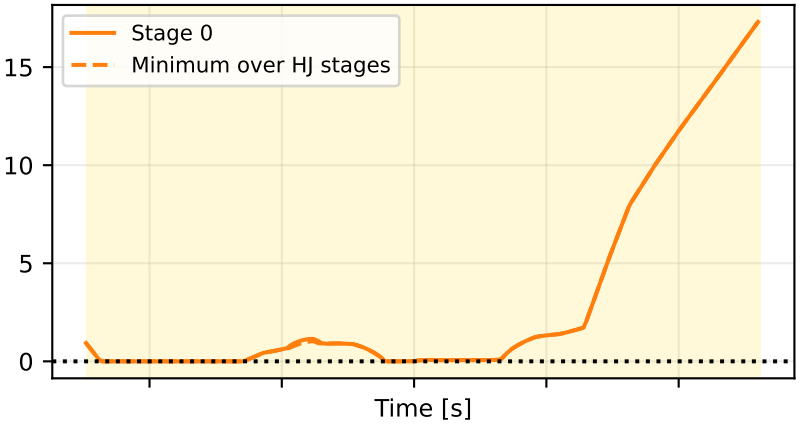
Lower slack



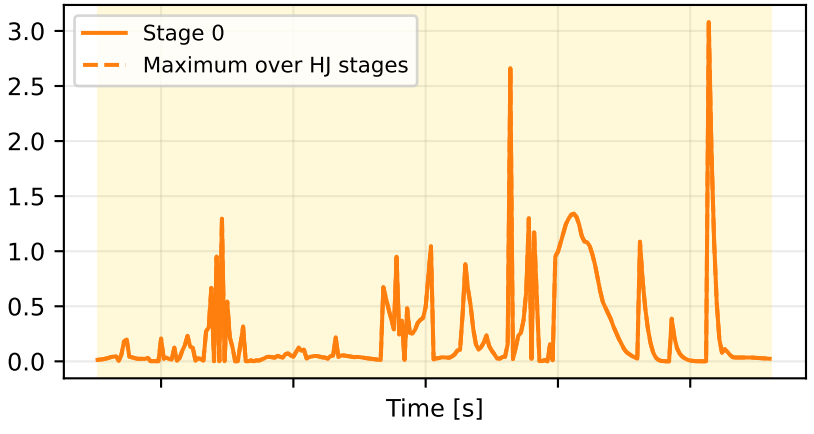
dce: BRT



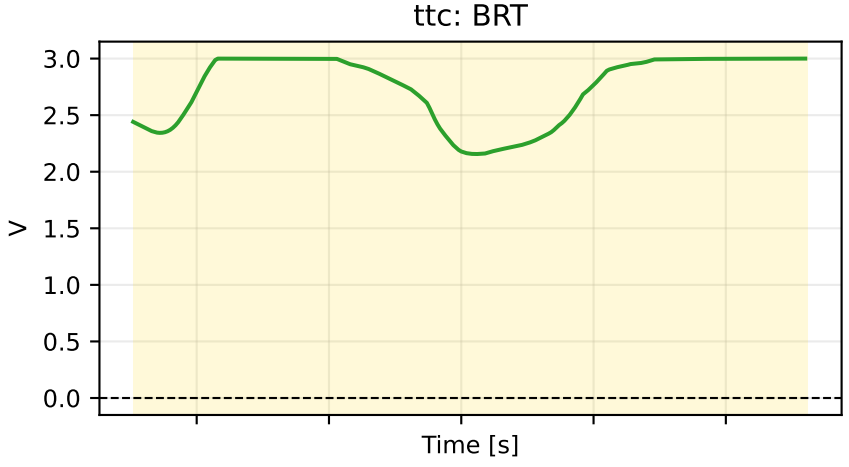
HJ residual



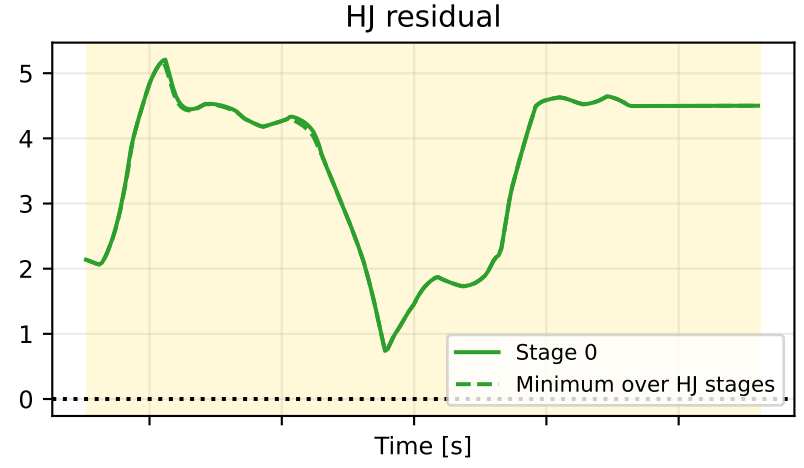
Lower slack



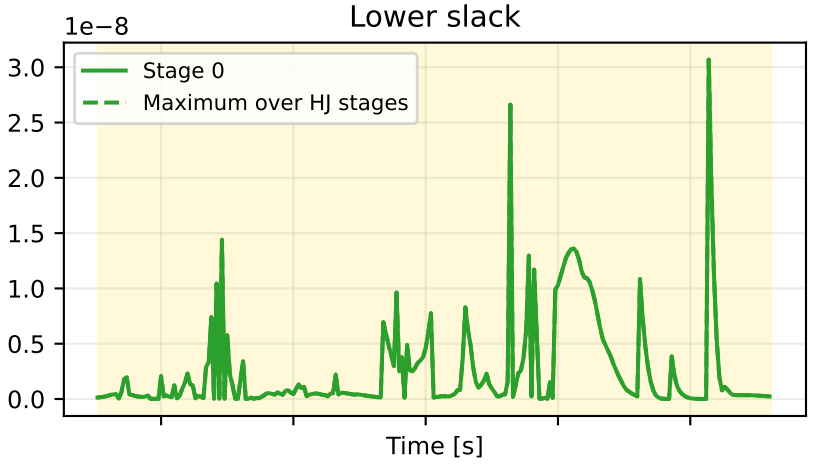
ttc: BRT



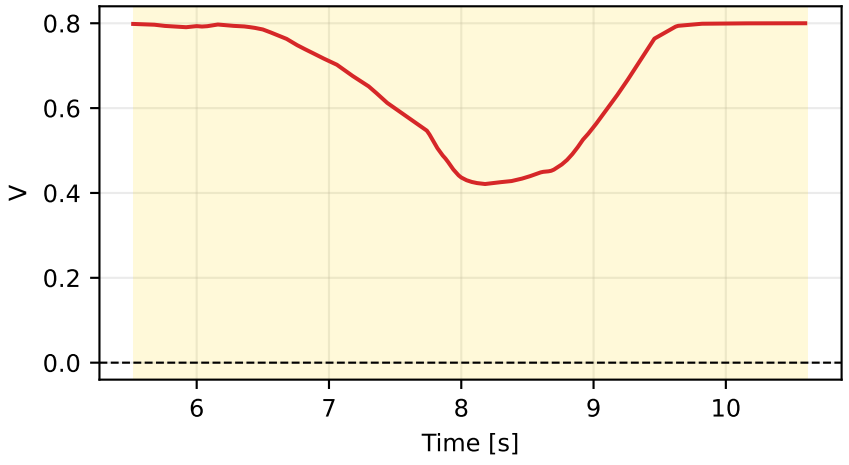
HJ residual



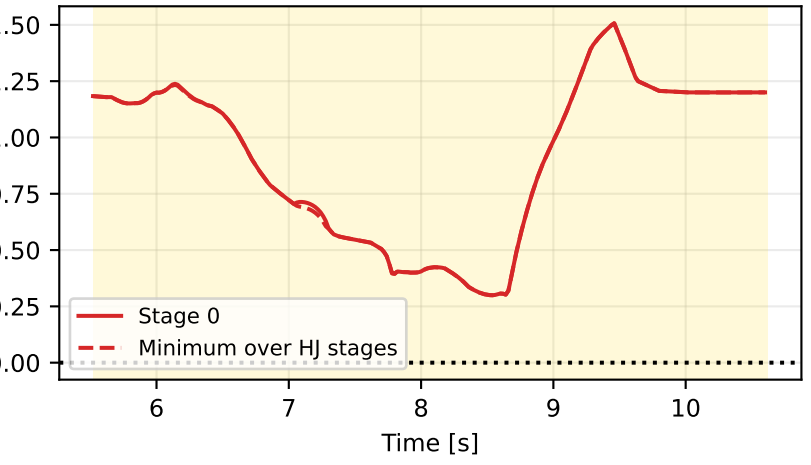
Lower slack



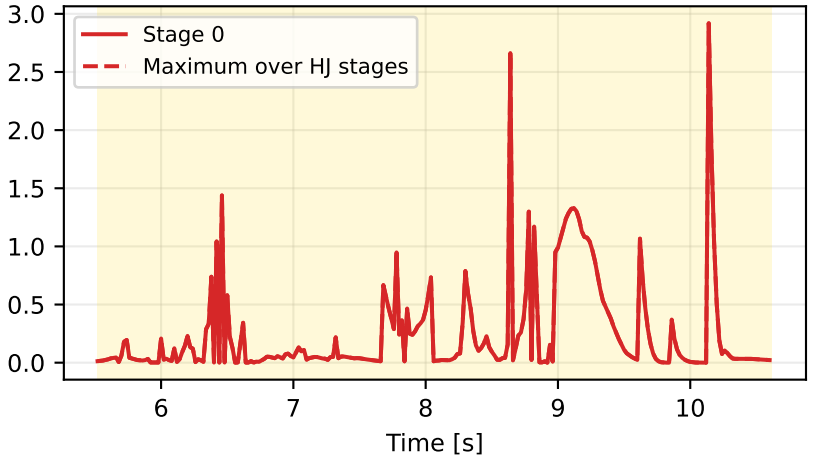
eggert: BRT

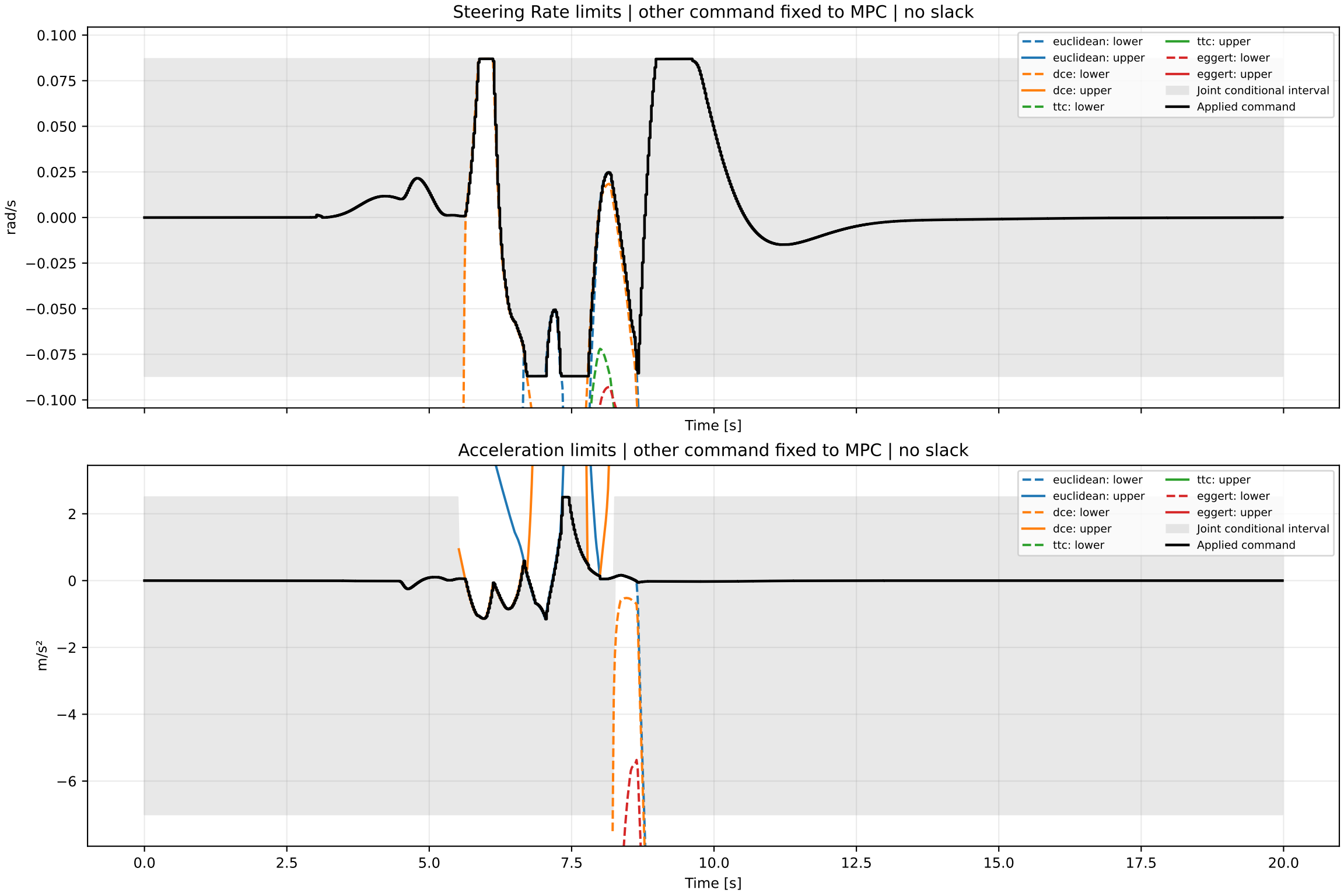


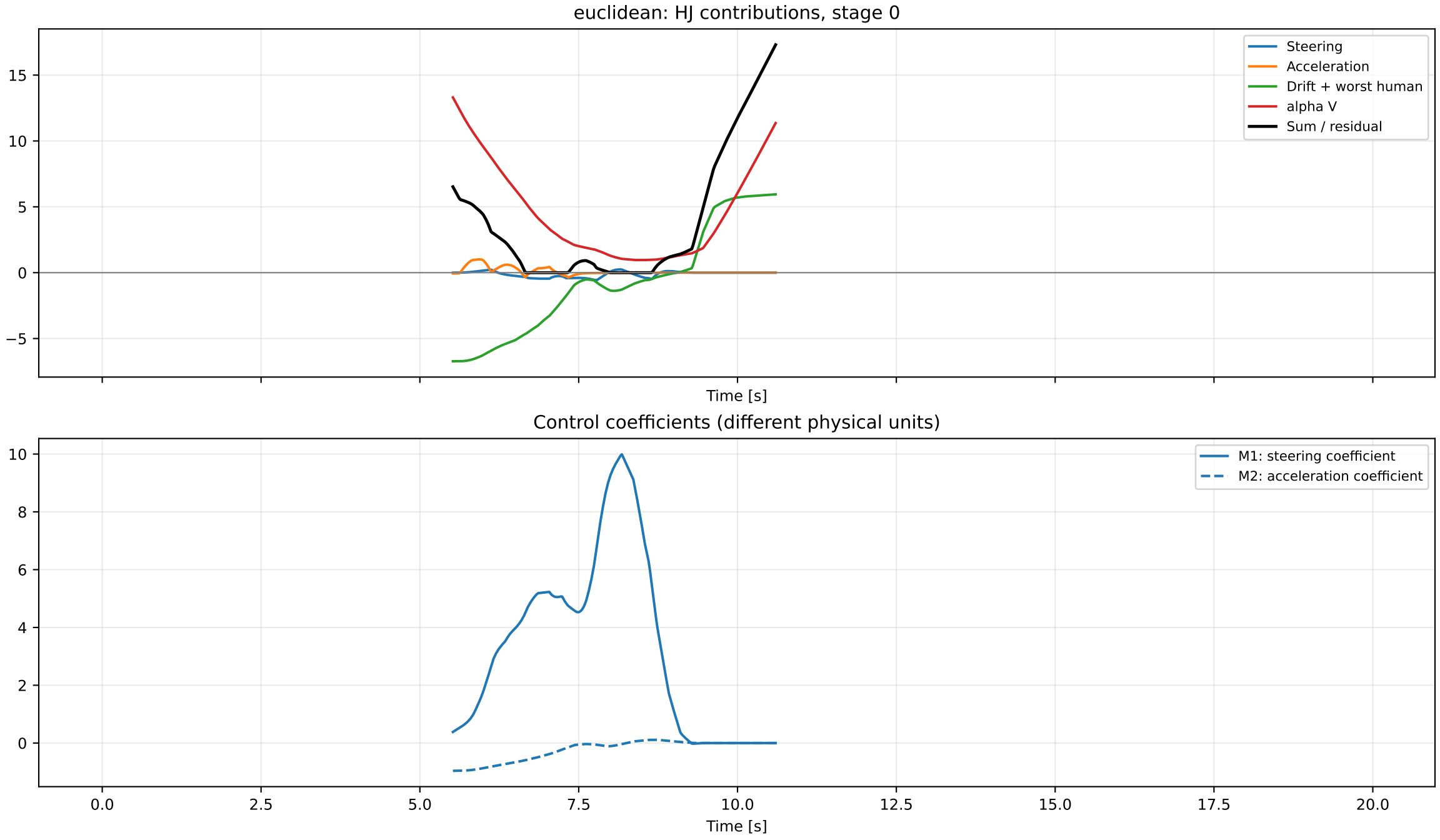
HJ residual

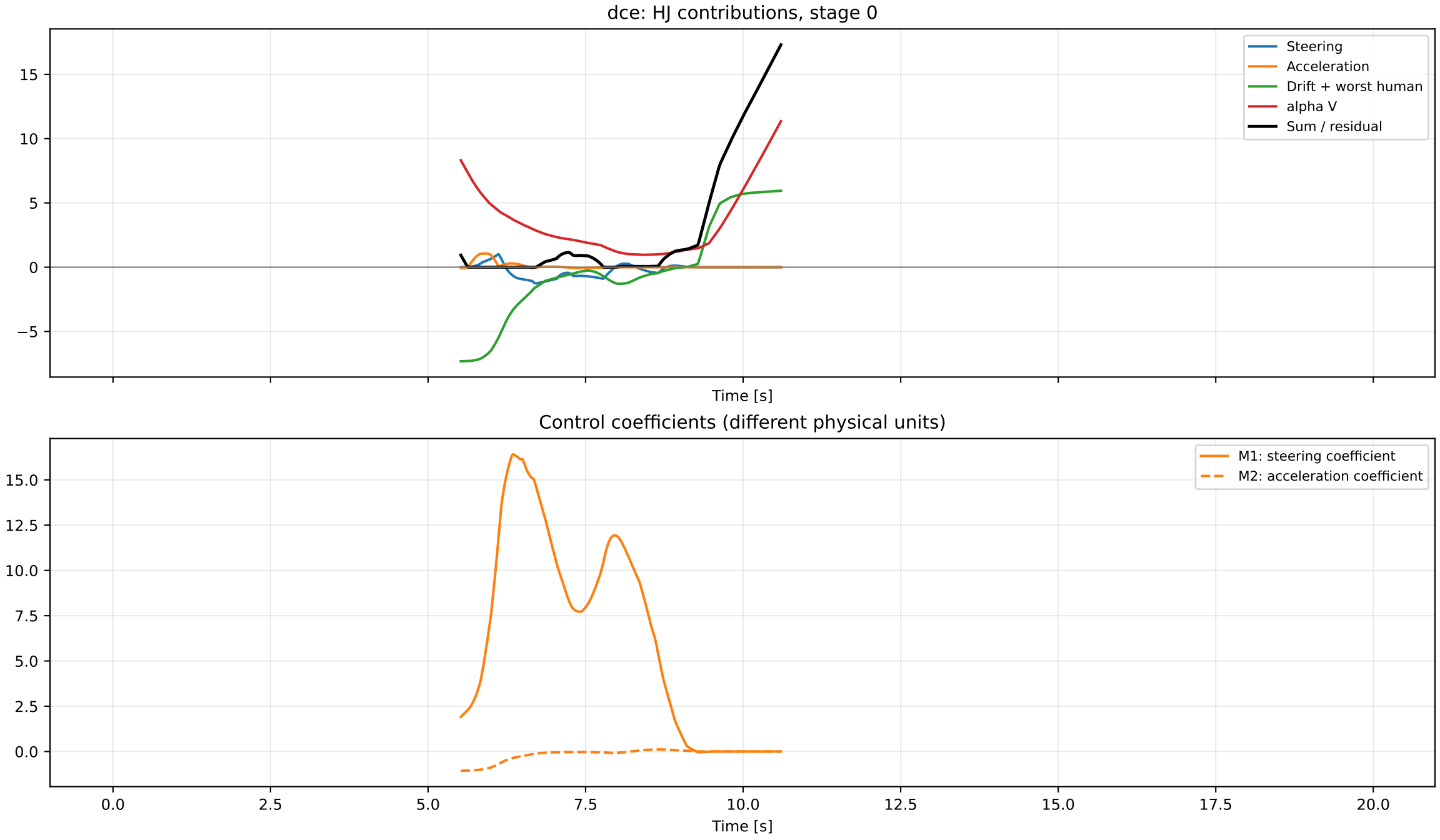


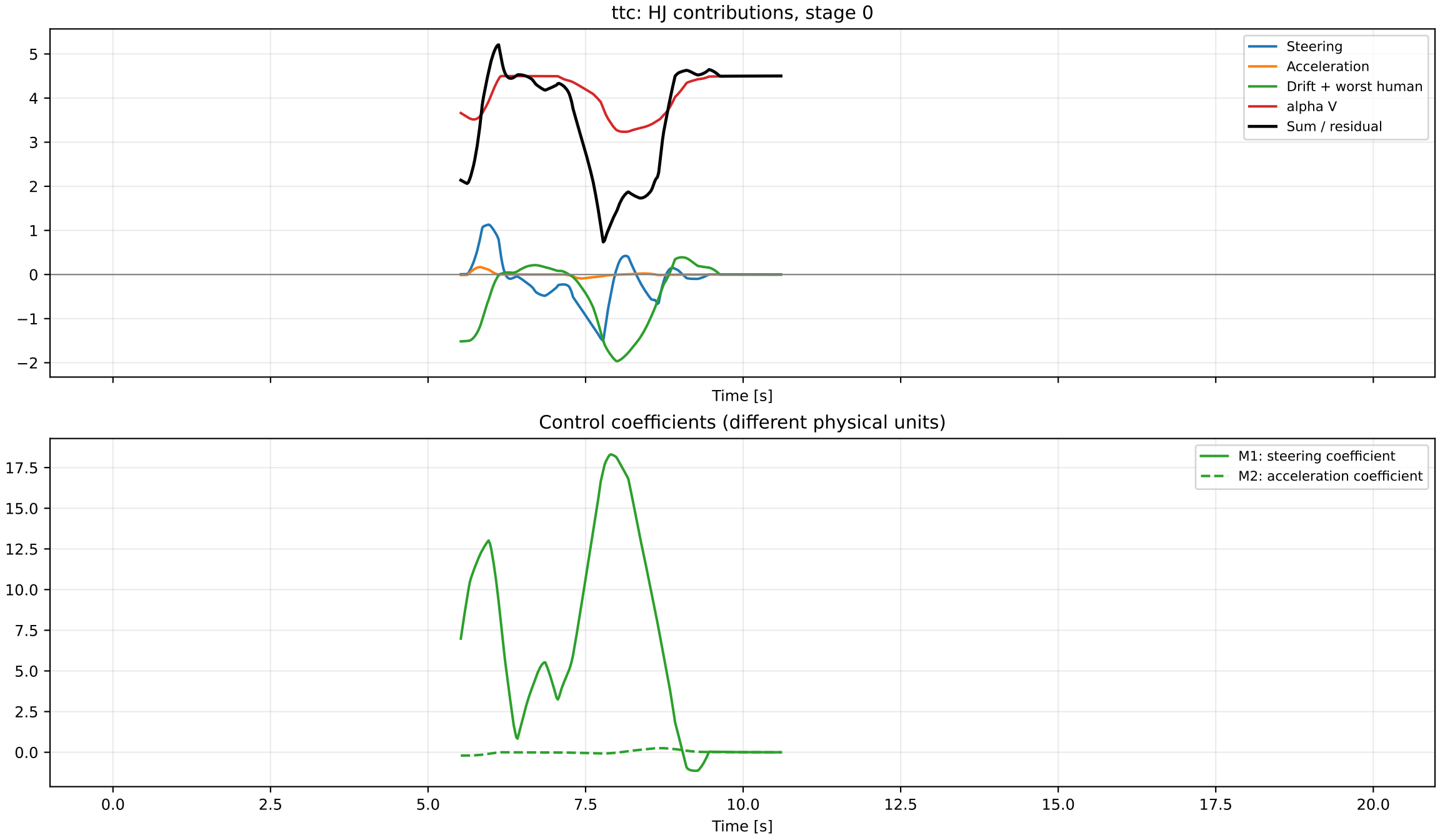
Lower slack

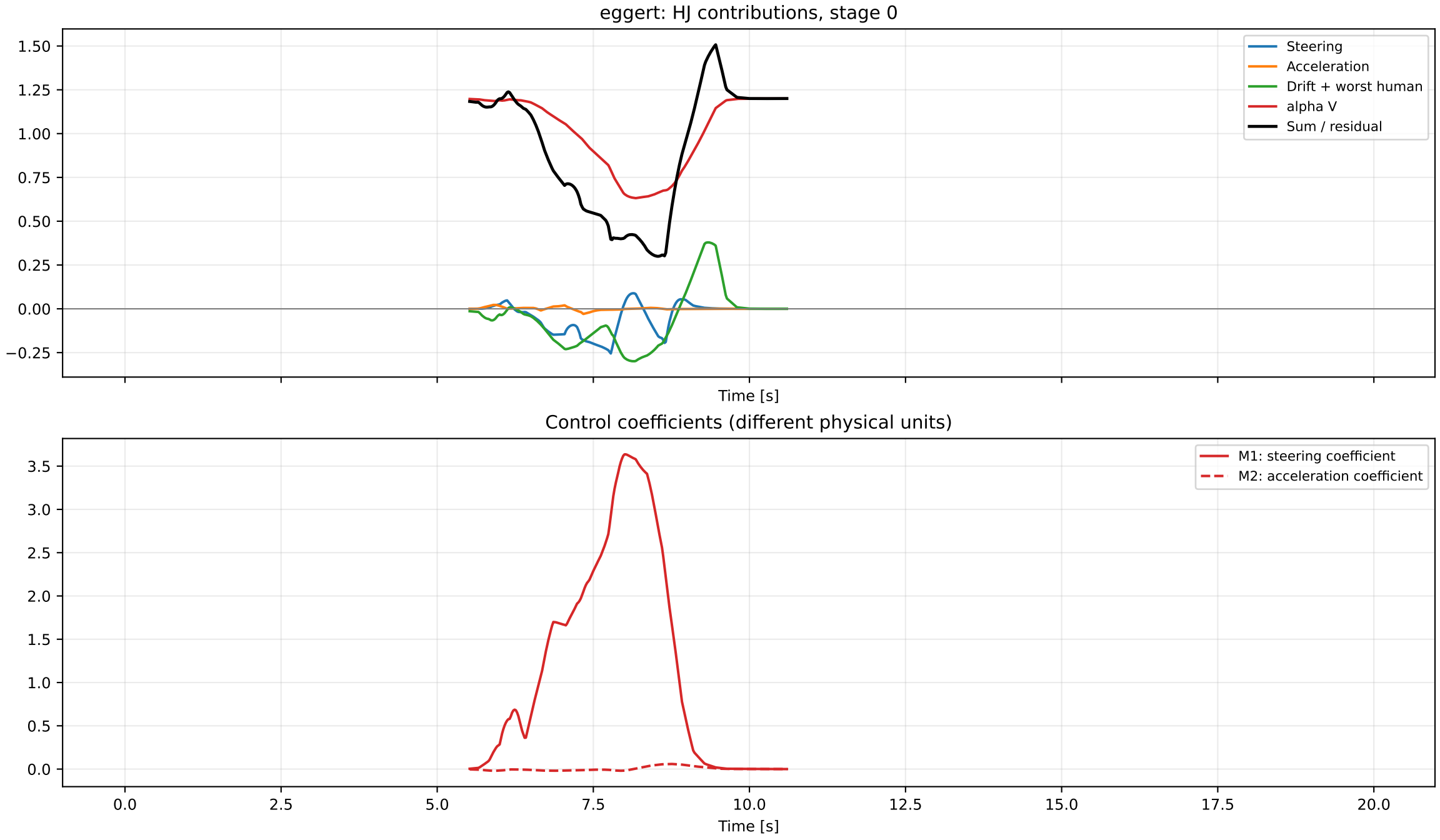


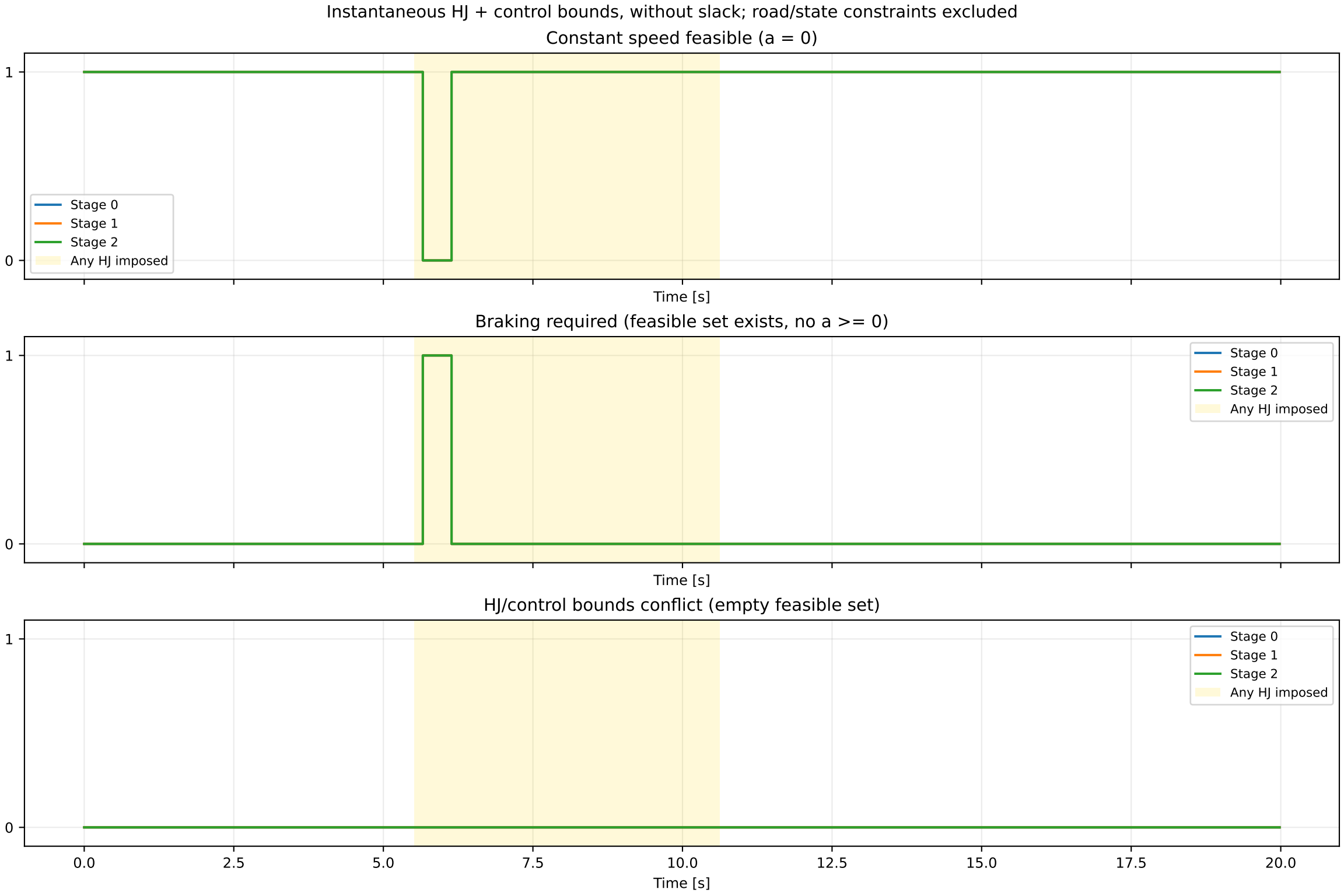

Steering Rate limits | other command fixed to MPC | no slack
rad/s
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
Time [s]
euclidean: lower
euclidean: upper
dce: lower
dce: upper
ttc: lower
ttc: upper
eggert: lower
eggert: upper
Joint conditional interval
Applied command
Acceleration limits | other command fixed to MPC | no slack
m/s²
2
0
−2
−4
−6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
euclidean: lower
euclidean: upper
dce: lower
dce: upper
ttc: lower
ttc: upper
eggert: lower
eggert: upper
Joint conditional interval
Applied command

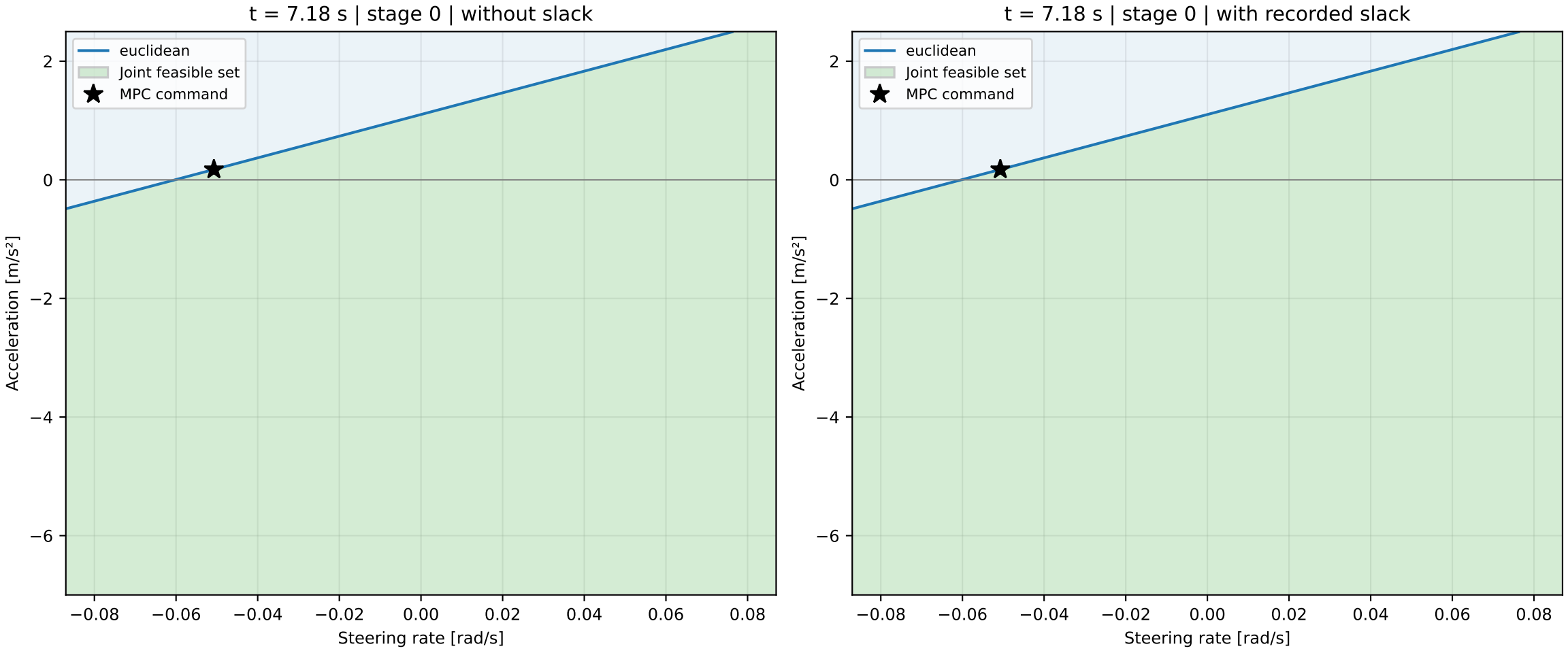

euclidean: HJ contributions, stage 0
Steering
Acceleration
Drift + worst human
alpha V
Sum / residual
15
10
5
0
−5
Time [s]
Control coefficients (different physical units)
M1: steering coefficient
M2: acceleration coefficient
10
8
6
4
2
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

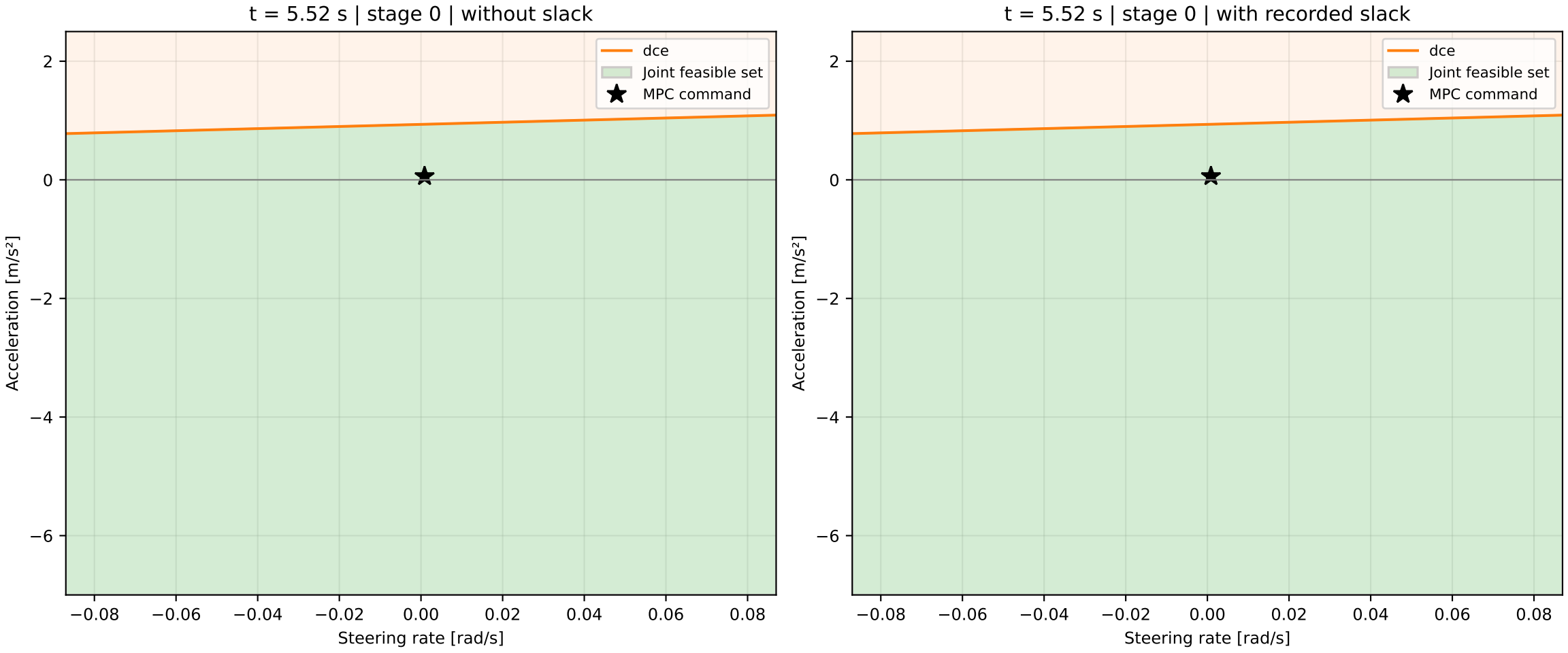

dce: HJ contributions, stage 0
Steering
Acceleration
Drift + worst human
alpha V
Sum / residual
15
10
5
0
−5
Time [s]
Control coefficients (different physical units)
M1: steering coefficient
M2: acceleration coefficient
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

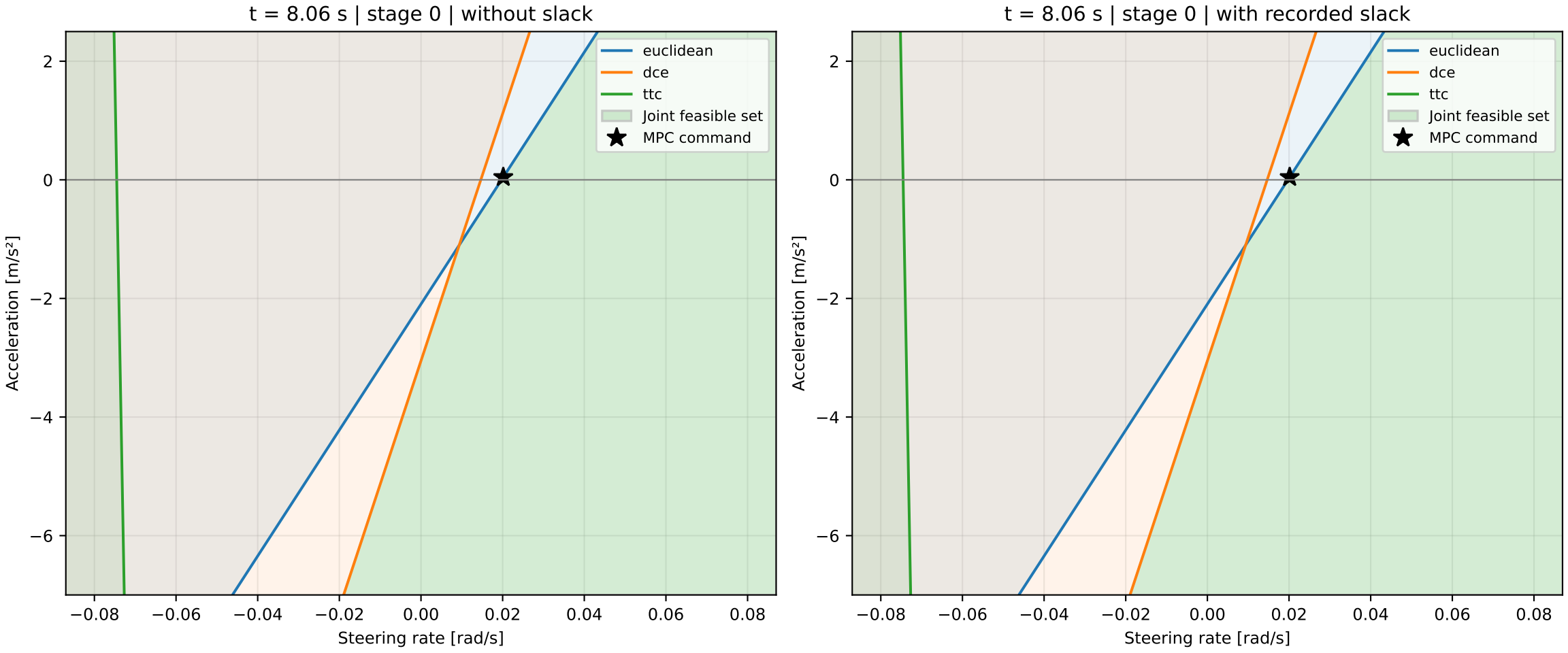

ttc: HJ contributions, stage 0
Steering
Acceleration
Drift + worst human
alpha V
Sum / residual
5
4
3
2
1
0
−1
−2
Time [s]
Control coefficients (different physical units)
M1: steering coefficient
M2: acceleration coefficient
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

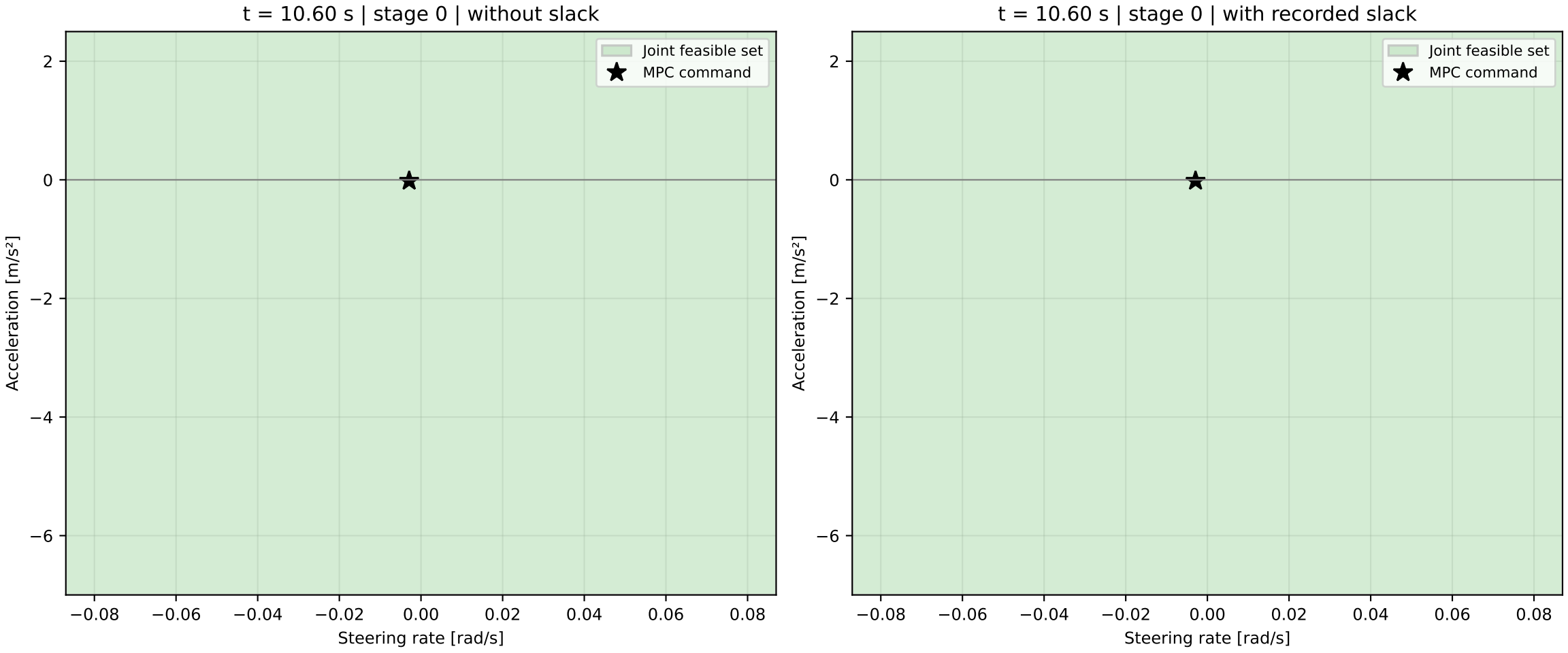

eggert: HJ contributions, stage 0
Steering
Acceleration
Drift + worst human
alpha V
Sum / residual
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
Time [s]
Control coefficients (different physical units)
M1: steering coefficient
M2: acceleration coefficient
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

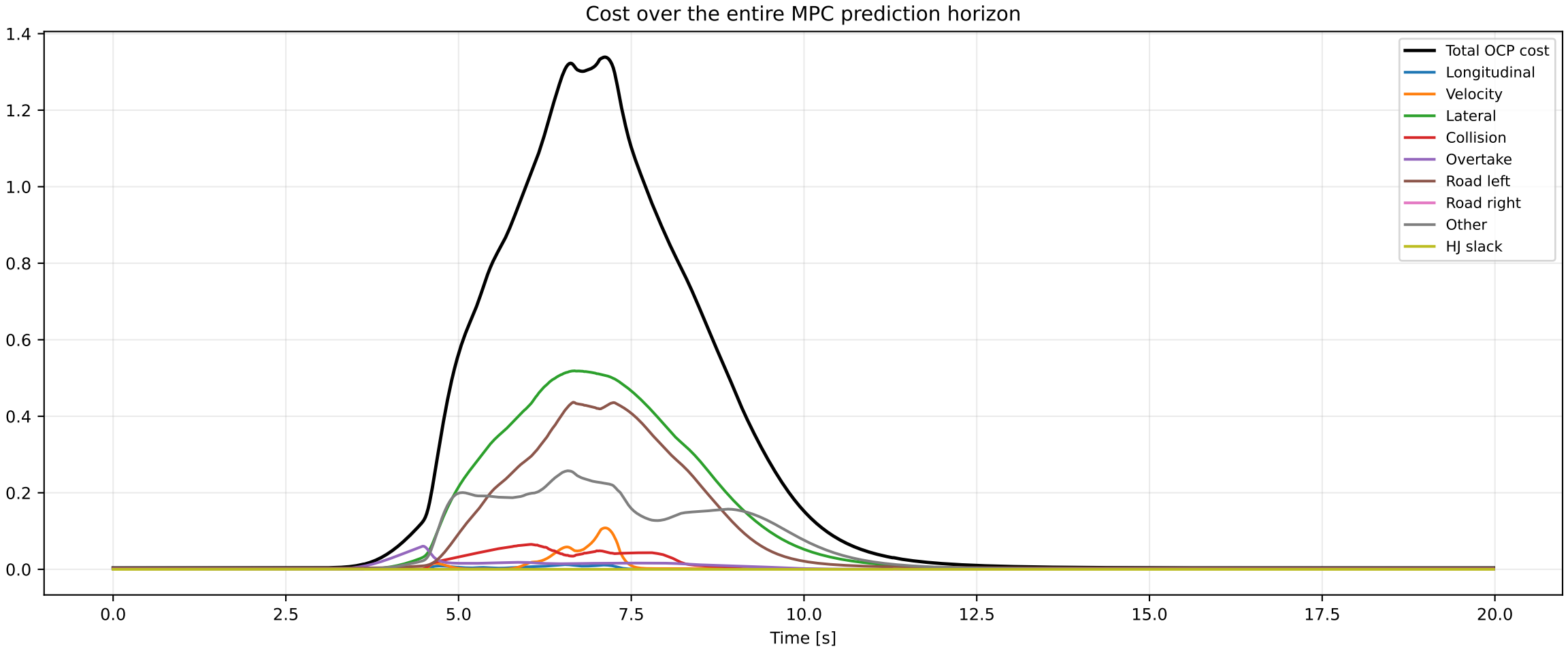

Instantaneous HJ + control bounds, without slack; road/state constraints excluded
Constant speed feasible (a = 0)
1
0
Stage 0
Stage 1
Stage 2
Any HJ imposed
Time [s]
Braking required (feasible set exists, no a >= 0)
1
0
Stage 0
Stage 1
Stage 2
Any HJ imposed
Time [s]
HJ/control bounds conflict (empty feasible set)
1
0
Stage 0
Stage 1
Stage 2
Any HJ imposed
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


t = 7.18 s | stage 0 | without slack
t = 7.18 s | stage 0 | with recorded slack
euclidean
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


t = 5.52 s | stage 0 | without slack
t = 5.52 s | stage 0 | with recorded slack
dce
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


t = 8.06 s | stage 0 | without slack
t = 8.06 s | stage 0 | with recorded slack
euclidean
dce
ttc
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


t = 10.60 s | stage 0 | without slack
t = 10.60 s | stage 0 | with recorded slack
Joint feasible set
MPC command
Acceleration [m/s²]
Steering rate [rad/s]
2
0
−2
−4
−6
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08


Cost over the entire MPC prediction horizon
Total OCP cost
Longitudinal
Velocity
Lateral
Collision
Overtake
Road left
Road right
Other
HJ slack
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]